An explicit three-player NOF exactly-N protocol with improved leading communication constant

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Editorial note. Editable transcription of the official CC BY 4.0 eight-page publication. Complete original Section2 proof retained; model and main theorem from Section1, direct Section3 consequence retained as context. Original entropy formulas, carry inequalities, cover weights and q/d choices are transcribed without mathematical correction or added assumptions. Standard background historical discussion and conjectures omitted. Bracket citation numbers retain original 23-entry bibliography numbering. Companion arxiv/journal identities denote the same result and are not counted again. All production and validation on Yupeng; compilation belongs to root.

Allowed standard prerequisites (editorial).

- NOF blackboard communication, conditional Shannon entropy, standard integral versus fractional set-cover bound. Official Sections1–2; standard Lovasz set-cover theorem cited as [15] with complete source-local use included

Model and conventions (original Section 1)

The multiplayer Number On the Forehead (NOF) model of communication complexity was introduced by Chandra, Furst and Lipton [9]. Given a function $f : [N]^k \to \{0,1\}$, the k players in this scenario should jointly find out $f(x_1,\ldots,x_k)$. We think of x_i as being placed on player i's forehead, so that each player sees the whole input bar one argument. Players communicate by writing bits on a shared blackboard according to an agreed-upon protocol. The exactly-N function $f : [N]^k \to \{0,1\}$ is given by $f(x_1,\ldots,x_k) = 1$ iff $\sum x_i = N$. All logarithms in this paper are in base 2. Also, unless otherwise specified, all asymptotic statements are taken with $N \to \infty$.

Theorem 1. *There is an explicit protocol for NOF exactly-N of complexity*

$$2\sqrt{\log e}\sqrt{\log N} + o(\sqrt{\log N}) = 2.4022\ldots\sqrt{\log N} + o(\sqrt{\log N}). \qquad (2)$$

Proof of Theorem 1 (original Section 2)

The three players in our protocol are called P_x, P_y and P_z. The inputs that they get to see are (y,z), (x,z) and (x,y) respectively. Here is a similar problem

in the realm of vector addition. Given integers $q, d > 1$, define $g = g_{q,d}(\alpha, \beta, \gamma)$ to be 1 if $\alpha + \beta = \gamma$ and 0 otherwise. Here $\alpha, \beta \in [q]^d$, $\gamma \in [2q]^d$ and addition is vector addition in $\mathbb{R}^d$. The following one-round protocol [3] for g is correct because the inequality $\|2\alpha - \gamma\|^2 + \|2\beta - \gamma\|^2 \geq 2\|\alpha - \beta\|^2$ holds always and is an equality iff $\gamma = \alpha + \beta$.

Protocol 1. A protocol for $g_{q,d}$.

1. P_z computes $\|\alpha - \beta\|_2^2$, and writes the result on the board.
2. P_y writes 1 iff $\|\alpha - \beta\|_2^2 = \|2\alpha - \gamma\|_2^2$.
3. P_x writes 1 iff $\|\alpha - \beta\|_2^2 = \|2\beta - \gamma\|_2^2$.

The cost of this protocol is $2 + \log dq^2$. The above is an efficient method to decide high-dimensional vector addition, but our objective is to decide the integer addition relation $X + Y + Z = N$. We let $x = X$, $y = Y$ and $z = N - Z$, so the relation we need to consider is $x + y = z$. Our protocol to decide whether $x + y = z$ builds on the protocol for $g_{q,d}$. It is the issue of carry bits in integer addition that makes this decision problem harder. The integers $q, d > 1$ are chosen so that

$$2qN > q^d \geq 2N. \tag{3}$$

the specific choice is made below so as to minimize the cost of the protocol. We denote by w_q the vector that corresponds to the base q representation of the integer w. As usual, e_i is the d-dimensional vector with 1 in the i-th coordinate and zeros elsewhere. Let $C(x, y) \in \{0, 1\}^d$ be the carry vector when x and y are added in base q. The relation $x + y = z$ among integers is equivalent to the vector relation

$$x_q + y_q = \zeta,$$

where the i-th coordinate of ζ is

$$\zeta_i = z_i + q \cdot C(x, y)_i - C(x, y)_{i-1}$$

(Here $C(x, y)_0 = 0$). The protocol from [3] now suggests itself: P_z posts $C(x, y)$, and Protocol 1 is used to decide the relation $x_q + y_q = \zeta$. This yields again the estimate (1) of the original paper, $2\sqrt{2}\sqrt{\log N} + o(\sqrt{\log N})$. The alternative approach that we adopt here considers instead the equivalent vector relation

$$x_q + \eta = z_q$$

where

$$\eta = (x + y)_q - x_q.$$

Concretely, the i-th coordinate of η is:

$$\eta_i = y_i - q \cdot C(x, y)_i + C(x, y)_{i-1}.$$

In order to run Protocol 1, P_z needs to know η and x_q, which he does. The situation with P_y is even simpler, since he needs to know x_q and z_q which are his inputs. The only difficulty is with P_x who needs to know z_q (which he does) and η. The latter is not part of his input and P_z fills in the missing information for him. The obvious solution is for P_z to reveal C to P_x using

d bits of information. However, we can save communication by exploiting the fact that P_x and P_z share some information, i.e., they both know y for every $y \neq 0$. By a standard argument in this area which we detail below (Proposition 2), a protocol that works for typical pairs x, y can be easily modified to work in all cases. So, let us pick x and y uniformly at random from among the d-digit numbers in base q and think of C, the vector of carry bits as a random variable on this probability space. The number of bits that P_z needs to post so that P_x gets to know C, and therefore know η, is $H(C \mid y)$, the entropy of C given y. The gain is clear, since $H(C) \geq H(C \mid y)$. It remains to estimate $H(C \mid y)$. Fix some integers $s \geq t \geq 0$, and let X be the random variable that is a uniformly sampled subset of $[s]$ of cardinality $\geq t$. It is easily verified that $H(X) = (1 + o_s(1)) \cdot s \cdot h(t/s)$, where $h(\cdot)$ is the univariate entropy function. The entropy of X is the same also if we sample subsets of $[s]$ of cardinality $< t$. Let r be an integer in the range $d \gg r \gg 1$, e.g., $r \approx \sqrt{d}$. For $j = 1, \ldots, r$, let

$$S_j = \{i \mid \tfrac{qj}{r} > y_i \geq \tfrac{q(j-1)}{r}\},$$

where $q > y_i \geq 0$ is the i-th digit of y. A carry occurs in digit $i \in S_j$ only if $x_i > \frac{q(r-j)}{r}$, where x_i is the i-th digit of x. Then

$$H(C \mid y) \leq (1 + o_r(1)) \sum_{j=1}^{r} \frac{|S_j|}{d} h(\tfrac{j}{r}).$$

Since y is chosen at random, $|S_j| \leq (1 + o_r(1))\frac{d}{r}$, and so

$$H(C \mid y) \leq (1 + o_r(1)) \sum_{j=1}^{r} \frac{1}{r} h(\tfrac{j}{r}).$$

The limit of this expression as $r \to \infty$ is

$$\lambda = \int_0^1 h(u)\, du = \frac{\log e}{2} = 0.721\ldots$$

It is left to optimize on q and d. The complexity of our protocol is

$$\lambda d + \log dq^2 + 2,$$

where recall that $2qN > q^d \geq 2N$. It is not hard to verify that choosing

$$d = \sqrt{\frac{2}{\lambda} \log 2N} \qquad q = 2^{\sqrt{\frac{\lambda}{2} \log 2N}}, \tag{4}$$

we get a protocol with complexity

$$2\sqrt{2\lambda \log N} + o(\sqrt{\log N}),$$

and this is asymptotically optimal in our setting. To sum up, here is the protocol which proves Theorem 1:

Protocol 2. A protocol for exactly-N, for typical pairs x, y. For d, q as in Equation (4)

1. P_z publishes the vector $\eta = (x+y)_q - x_q$ in a way that P_x can read.
2. The players run protocol 1 for $g_{q,d}$ on inputs x_q, η, z_q. That is:
   (a) P_z writes $\|\eta - x_q\|_2^2$ on the board
   (b) P_y writes 1 iff $\|\eta - x_q\|_2^2 = \|2x_q - z_q\|_2^2$.
   (c) P_x writes 1 iff $\|\eta - x_q\|_2^2 = \|2\eta - z_q\|_2^2$.

Proposition 2. *Let P be an NOF protocol for the exactly-N that works correctly for an $\Omega(1)$-fraction of the input pairs x, y (and every z) with communication complexity $\Phi(N)$. Then there is an NOF protocol that works for all inputs with communication complexity $\Phi(N) + O(\log\log N)$.*

Proof. Let $S \subseteq [N] \times [N]$ be the set of input pairs x, y on which P succeeds. We claim that there is a collection F of $O(\log N)$ vectors $\Delta \in [N] \times [N]$ such that

$$\bigcup_{\Delta \in F} (S + \Delta) \supseteq [N] \times [N].$$

In the modified protocol, P_z sees x, y and announces the index of some $\Delta = (\Delta_1, \Delta_2) \in F$ for which $(x - \Delta_1, y - \Delta_2) \in S$. Then the players run Protocol 2 with inputs $(x - \Delta_1, y - \Delta_2, z - \Delta_1 - \Delta_2)$. The construction of F uses a standard fact about the set-cover problem. For a family of finite sets $\mathcal{X} \subseteq 2^\Omega$ we denote by $c(\mathcal{X})$ the least number of members in $\mathcal{X}$ whose union is Ω. Also $c^*(\mathcal{X})$ is the minimum cost of a fractional cover. Namely,

$$c^*(\mathcal{X}) = \min \sum_{\mathcal{X}} \omega_X, \quad \text{where } \omega_X \geq 0 \text{ for every } X \in \mathcal{X} \text{ and } \sum_{x \in X} \omega_X \geq 1 \text{ for every } x \in \Omega.$$

Then

$$c(\mathcal{X}) \leq \log(|\Omega|) \cdot c^*(\mathcal{X})$$

(e.g., Lovász [15]) and actually the greedy algorithm yields a set cover that meets this bound. In our case $\Omega = [N] \times [N]$, and

$$\mathcal{X} = \{(S + \Delta) \cap ([N] \times [N]) \mid \Delta \in [-N, N] \times [-N, N]\}.$$

It is easily verified that the weights $\omega_x = \frac{10}{N^2}$ constitute a fractional cover, so that $c^*(\mathcal{X}) \leq 40$ and hence $c(\mathcal{X}) \leq 80 \log N$, as claimed. □

Applications in additive combinatorics (original Section 3, contextual consequence)

Let $\rho_k(N)$ be the largest density of a subset of $[N]$ without an arithmetic progression of length k. A corner is a triple of points $(x, y), (x + \delta, y), (x, y + \delta)$ for some $\delta \neq 0$. Denote the largest density of a subset of $[N] \times [N]$ without a corner by $\rho_3^{\perp}(N)$. As mentioned in the introduction, the NOF communication complexity of f is closely related to corners theorems. Our Theorem 1 immediately implies,

Theorem 3.

$$\rho_3^{\angle}(N) \geq 2^{-2\sqrt{\log e}\sqrt{\log N}+o(\sqrt{\log N})} = 2^{-2.4022...\sqrt{\log N}+o(\sqrt{\log N})}.$$

There is an explicit corner-free subset of $[N] \times [N]$ *of size*

$$N^2/2^{2\sqrt{\log e}\sqrt{\log N}+o(\sqrt{\log N})}.$$

The derivation of Theorem 3 from Theorem 1 is an easy consequence of the following claim.

Claim 4 ([9], implicit)**.** *1. There is an optimal one-round protocol for the addition problem.*

2. Let $T = T(x, y)$ *be the message that the* P_z *sends on inputs* (x, y) *in a one-round protocol for the addition problem. Then the set*

$$S(T) = \{(x, y) : T(x, y) = T\}$$

is corner-free.

See [9,7,1,14,3] for more details about the above claim and the relation between communication complexity and additive combinatorics. The same comments and corollaries above apply also verbatim to the $(6, 3)$ Theorem (e.g., [19]) and to the quantitative version of the triangle removal lemma.

References

[1] A. Ada, A. Chattopadhyay, O. Fawzi, and P. Nguyen. The NOF multiparty communication complexity of composed functions. computational complexity, 24(3):645–694, 2015.

[2] M. Ajtai and E. Szemerédi. Sets of lattice points that form no squares. Stud. Sci. Math. Hungar, 9(1975):9–11, 1974.

[3] N. Alon and A. Shraibman. Number on the forehead protocols yielding dense ruzsa–szemerédi graphs and hypergraphs. Acta Mathematica Hungarica, 161(2):488–506, 2020.

[4] P. Beame, M. David, T. Pitassi, and P. Woelfel. Separating deterministic from randomized nof multiparty communication complexity. In Proceedings of the 34th International Colloquium On Automata, Languages and Programming, Lecture Notes in Computer Science. Springer-Verlag, 2007.

[5] P. Beame, T. Pitassi, and N. Segerlind. Lower bounds for Lovász-Schrijver systems and beyond follow from multiparty communication complexity. SIAM Journal on Computing, 37(3):845–869, 2006.

[6] F. A. Behrend. On sets of integers which contain no three terms in arithmetical progression. Proceedings of the National Academy of Sciences, 32(12):331–332, 1946.

[7] R. Beigel, W. Gasarch, and J. Glenn. The multiparty communication complexity of Exact-T: Improved bounds and new problems. In International Symposium on Mathematical Foundations of Computer Science, pages 146–156. Springer, 2006.

[8] T. F. Bloom and O. Sisask. Breaking the logarithmic barrier in Roth's theorem on arithmetic progressions. arXiv preprint, 2020. arXiv:2007.03528.

[9] A. Chandra, M. Furst, and R. Lipton. Multi-party protocols. In Proceedings of the 15th ACM Symposium on the Theory of Computing, pages 94–99. ACM, 1983.

[10] M. Elkin. An improved construction of progression-free sets. In Proceedings of the twenty-first annual ACM-SIAM symposium on Discrete Algorithms, pages 886–905. Society for Industrial and Applied Mathematics, 2010.

[11] W. T. Gowers. Hypergraph regularity and the multidimensional szemerédi theorem. Annals of Mathematics, pages 897–946, 2007.

[12] J. Håstad and M. Goldmann. On the power of small-depth threshold circuits. Computational Complexity, 1:113–129, 1991.

[13] E. Kushilevitz and N. Nisan. Communication Complexity. Cambridge University Press, 1997.

[14] N. Linial, T. Pitassi, and A. Shraibman. On the communication complexity of high-dimensional permutations. In 10th Innovations in Theoretical Computer Science Conference, ITCS San Diego, California, USA, volume 124, pages 54:1–54:20, 2019.

[15] L. Lovász. On the ratio of optimal integral and fractional covers. Discrete Mathematics, 13:383–390, 1975.

[16] B. Nagle, V. Rödl, and M. Schacht. The counting lemma for regular k-uniform hypergraphs. Random Structures & Algorithms, 28(2):113–179, 2006.

[17] V. Rödl and J. Skokan. Regularity lemma for k-uniform hypergraphs. Random Structures & Algorithms, 25(1):1–42, 2004.

[18] K. F. Roth. On certain sets of integers. Journal of the London Mathematical Society, 1(1):104– 109, 1953.

[19] I. Ruzsa and E. Szemerédi. Triple systems with no six points carrying three triangles. Combinatorics (Keszthely, 1976), Coll. Math. Soc. J. Bolyai, 18:939–945, 1978.

[20] J. Solymosi. Note on a generalization of Roth's theorem. Discrete and Computational Geometry: The Goodman-Pollack Festschrift, pages 825–827, 2003.

[21] E. Szemerédi. On sets of integers containing no k elements in arithmetic progression. Acta Arith, 27(199-245):2, 1975.

[22] B. L. van der Waerden. Beweis einer Baudetschen Vermutung. Nieuw Arch. Wiskunde, 15:212–216, 1927.

[23] A. Yao. On ACC and threshold circuits. In Proceedings of the 31st IEEE Symposium on Foundations of Computer Science, pages 619–627. IEEE, 1990.